Gokul Raju Govinda Raju

Robotics Research Engineer

Robotics, Systems and Control MSc at ETH Zurich

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EDUCATION

ETH Zurich — Zurich, Switzerland
MSc in Robotics, Systems and Control — Sep 2022 –Aug 2025

- **Master Thesis:** State-Space Models for Efficient Reinforcement Learning in Quadrotors
- **Coursework:** Machine Learning, Probabilistic AI, Computer Vision, 3D Vision, Planning & Decision Making for Autonomous Robots, Robot Dynamics, Dynamic Programming & Optimal Control, Algorithms & Fairness

Heriot-Watt University — Dubai, UAE
BEng in Mechanical Engineering (Hons.) — Sep 2018 –Aug 2021

- **Bachelor Thesis:** Fault Diagnosis of Rolling Element Bearings using Artificial Neural Networks
- **Watt Club Medal:** Awarded to the student achieving **University Rank: 1/254**

SKILLS

Programming: Python, C++, MATLAB

Frameworks & Tools: PyTorch, JAX, TensorFlow, Keras, NumPy, SciPy, scikit-learn, Matplotlib, Pandas, HuggingFace

Simulation & Perception Tools: IsaacSim, IsaacLab, OpenCV

MLOps: Weights & Biases, ONNX, TensorRT, PyTorch DDP, AWS, Git, Docker, Kubernetes, SQL, FastAPI, Flask

Languages: English (C2), Tamil (C2), Hindi (C1)

EXPERIENCE

FPrime AI — Zurich, Switzerland
Machine Learning Engineer (Intern) — Apr 2025 –Feb 2026

- Fine-tuned **FLUX.1 (rectified-flow transformer model)** for product photography using PEFT techniques like **LoRA** with **quantization** for efficient training and inference
- Trained on **variable angles/lighting conditions** to enable controllable, detail-consistent image generation
- Developed **RAG (Retrieval-Augmented Generation)** assistants using Llama 3.2 and Qdrant to automate order queries and cross-selling, driving a **40% increase** in customer retention

Robotics and Perception Group, UZH — Zurich, Switzerland
Graduate Researcher — Aug 2023 –Oct 2024

- **Master Thesis:** State-Space Models for Efficient Reinforcement Learning in Quadrotors [GitHub]
- Proposed the first **SSM-based RL** framework for Autonomous Drone Racing for both state and vision based domains
- Developed a novel PPO JAX RL framework for Flightmare Simulator, resulting in SOTA performance - **10% faster lap times** and **50% faster inference times** compared to existing policies
- **Semester Thesis:** Pushing the Limits of Optical Flow Estimation for Event Cameras [GitHub]
- Pioneered the first **multi-event** optical flow framework for neuromorphic vision, achieving an **8% reduction in EPE**
- **Accelerated inference times by 30%** by optimizing the model architecture with a SSM-based encoder

Heriot-Watt University — Dubai, UAE
Research Assistant — Sep 2021 –Aug 2022

- Curated bearing-vibration datasets with over **1000 samples**, featuring varied fault sizes and RPM ranges
- Conducted ablations on spectrogram CNNs and achieved **97% classification accuracy** through hyper-parameter tuning
- Tutored **120** undergraduate students in CoppeliaSim and OnShape; delivered **6 labs** and **4 demos**

PUBLICATIONS

Perturbed State Space Feature Encoders for Optical Flow with Event Cameras Jun 2025
Gokul Raju Govinda Raju, Nikola Zubic, Marco Cannici, Davide Scaramuzza [arXiv]
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Nashville, 2025

PROJECTS

GPT-2 [GitHub]
Personal Project

- Implemented **GPT-2 (124M parameters) model** from scratch in PyTorch, including the the training pipeline
- Pretrained the model on ∼**10B tokens** using distributed training **(PyTorch DDP)** on a **4xRTX4090 GPU cluster**

Spaceship Obstacle Avoidance and Trajectory Planning [GitHub]
Planning and Decision Making for Autonomous Robots Project

- Implemented a **SCvx (Successive Convexification)** planning and control algorithm for a 2-D spaceship to avoid static obstacles (planets) & dynamic obstacles (satellites) and to dock with the docking station

Robust Soccer Ball Detection and Tracking (FIFA) [GitHub]
3D Vision Project

- Engineered a synthetic dataset with Google Research Football Simulator to enhance YOLOv8n object detection through transfer learning and improved **recall by 717%** and **precision by 216%**
- Integrated the fine-tuned detection model with a state-of-the-art approach, Motion-from-Blur (MfB) to handle high-speed blurry motion, resulting in a **8% improvement** in soccer ball detections
- Combined multi-view detections with camera pose information to triangulate the position of the soccer ball **(95.78% tracking accuracy at 0.75 IoU)** and visualized the tracking results by generating 3D trajectories for analytics

Fault Diagnosis of Rolling Element Bearings using Artificial Neural Networks
Bachelor Thesis

- Engineered an ANN-based fault diagnosis system using MATLAB's Wavelet and Neural Network Toolboxes, achieving **99.5% fault classification accuracy** in bearings
- Evaluated the system's performance on unseen data, achieving **98.64% accuracy** in fault magnitude estimation
- Implemented a data augmentation strategy to address class imbalance, leading to a **20% improvement** in minority class classification and enhanced model robustness

ATLAS-D: Industrial Autonomous Robot Vacuum
Industrial Project

- Collaborated with EGA (Emirates Global Aluminum) to design a functional CAD Design for an industrial autonomous robot vacuum to clean substations
- Designed a CAD model of the robot vacuum and an algorithm assisted by LiDAR Sensors and IR sensors on ROS, Gazebo, and RViz for mapping and navigation
- Designed a 4-stage dust separation system (multi-cyclone, HEPA-grade washable filters, BLDC-driven impeller) targeting a >**99% capture efficiency** for fine dust particles

HONORS AND AWARDS

UAE Golden Visa for Academic Excellence 2022–2032
James Anderson Memorial Prize (£5000) 2021
Eric Gibb Prize 2020
Deputy Principal's Award 2021, 2019
Heriot-Watt Merit Scholarship (AED 30000) 2018